ComplaintIQ: Complete Master Presentation Script & Course Policy Defense

Autonomous Financial Grievance Intelligence, Anomaly Detection & Time-Series Forecasting Platform

Candidate Name	Roll Number	SAP ID	Core Project Responsibility
Aryan Turalkar	A152	70562300045	NLP Feature Pipeline, Calibrated Linear SVM, Platt Scaling & Token Attribution
Suryansh Saraf	A125	70562300034	Full-Stack Architecture, React 18 / Tailwind Dashboard & FastAPI Microservices
Soham Hande	A129	70562300062	Seasonality-Aware Rolling MAD Anomaly Engine, Isolation Forest & NMF Topics
Pratham Deepak	A114	70562300078	Damped Additive Holt-Winters Forecasting, Alert Deduplication & State Machine

PROJECT BRIEF & EXECUTIVE SUMMARY

Executive Summary & Platform State: ComplaintIQ is a production-grade NLP and time-series decision support system operating over **6,924 real-world banking and financial complaints** across 8 core product verticals. It couples sublinear TF-IDF representation and Calibrated Linear SVM (**98.7% accuracy, 0.985 F1**) with white-box token-level feature attribution ($w_i \cdot x_i$), unsupervised Non-Negative Matrix Factorization (NMF, $k=8$), a Day-of-Week normalized Rolling Median Absolute Deviation (MAD) anomaly detector that eliminates 100% of weekend false alarms, and a Damped Holt-Winters forecasting engine yielding a 32% error reduction over seasonal baselines. This master script is your comprehensive reference guide for conducting the live web application walkthrough, defending academic Course Policy alignment, and acing the faculty cross-examination.

1. Live Presentation Setup, Timing & Protocol

Pre-Presentation Setup Checklist:
• **Active URLs:** Open http://localhost:5173 in Google Chrome or Safari in **Full Screen Mode**. Ensure the backend is healthy on http://localhost:8001/api/health (returning 6,924 complaints).
• **Clipboard Buffer:** Have this test string pre-copied in case you need to paste manually: I noticed an unauthorized charge of $450 from an unknown foreign merchant on my credit card that I never made.
• **Pacing Allocation:** Total presentation time is 10 to 12 minutes, structured as: Home Command Center (2 min) $\to$ Live AI Prediction Demo (2 min) $\to$ Complaint Explorer (1.5 min) $\to$ NLP Intelligence (1.5 min) $\to$ ML Intelligence & Topics (1.5 min) $\to$ Alert Center (1 min) $\to$ Conclusion & Team Roles (1 min) $\to$ Faculty Viva Cross-Examination (3–5 min).

2. Complete Word-for-Word Live-Frontend Speaking Script

The following script is designed for live screen-sharing. As you speak, follow the exact **[Action]** cues with your mouse cursor and clicks. Deliver the text in a natural, authoritative tone.

Step 1: Opening & Banking Domain Motivation [0:00 – 1:45]

[Screen Position]: Home Command Center (http://localhost:5173/) in full screen. Mouse hovering over header badge.
"Respected faculty members, good morning / afternoon. Today, rather than flipping through static presentation slides, our team will demonstrate our fully functional capstone platform live directly from the running web browser.

*Our project is **ComplaintIQ** — an Autonomous Financial Grievance Intelligence, Anomaly Detection & Time-Series Forecasting*

Platform.
*My team members are **Aryan Turalkar (A152)**, **Soham Hande (A-129)**, **Pratham Deepak (A-114)**, and myself, **Suryansh Saraf (A-125)**.*

***The Problem We Solved:** In modern retail and commercial banking, institutions receive thousands of grievances daily across Credit Cards, Savings Accounts, Home Mortgages, UPI & Net Banking, and Loans. In standard operations, human triage takes between **24 to 72 hours per record**, delaying critical fraud containment and risking statutory penalties from regulatory bodies like the RBI and CFPB. Furthermore, when systemic failures occur—such as an automated overdraft batch glitch or a regional UPI switch outage—operations teams only detect them days later because complaint volume drops 70% over weekends, blinding naive monitoring tools.*

*To solve this, we engineered an end-to-end intelligence system operating over **6,924 real-world banking complaints**, powered by FastAPI, React 18, TypeScript, TailwindCSS, Scikit-Learn, and SQLite."*

Step 2: Executive KPI Ribbon & 30-Day Volume Trajectory [1:45 – 3:15]

[Action]: Hover over the 4 Top KPI cards, then hover over the Recharts volume trend chart. Toggle 'Weekly', then switch back to 'Daily'. Point cursor to red circular anomaly markers.
*"Here on the **Home Command Center**, operations managers obtain real-time situational awareness:*
*• **Inbound Volume:** Tracks active complaint volume across our 6,924 corpus and percentage change versus the historical baseline.*
*• **Negative Sentiment Rate:** Real-time proportion of grievances expressing customer frustration or escalation threats.*
*• **Average Sentiment Score:** Bounded continuously on a calibrated -1.0 to +1.0 scale.*
*• **Active Warnings:** Displays currently flagged anomaly incidents requiring operational attention.*

*Right below is our **Interactive Volume Trajectory** built using Recharts. It plots daily customer complaint volume against moving averages. Notice these highlighted circular red markers: these are **not** random daily variances. Our backend statistical anomaly engine identified them as abnormal volume surges where complaint velocity significantly exceeded seasonal weekday thresholds. If an operations lead clicks on any spike, the system automatically cross-filters the underlying complaints in the Explorer for immediate investigation."*

Step 3: The Showstopper — Live AI Classifier & Feature Attribution [3:15 – 5:00]

[Action]: On the right-side card, click the sample chip '■ Unauthorized Card Charge'. The text auto-populates. Click 'Analyze Complaint'. Point to the result appearing in under 20 milliseconds.
*"Now, let us demonstrate our live model inference in real time. In this **Live AI Classifier Box**, an operations officer can test any customer dispute letter, email, or chatbot escalation.*
[Click Analyze Complaint]
*In **under 20 milliseconds**, the FastAPI backend executes our sublinear TF-IDF pipeline:*
*1. **Classification Output:** It classified this complaint as **'Unauthorized Fraud'** with **99.9% Calibrated Confidence**.*
*2. **Probability Distribution:** Below, you see normalized posterior probabilities across all 8 banking categories: Billing & Fees, Loan & Mortgage, Transaction Failure, Account & KYC, Credit Reporting, etc.*
*3. **Mathematical Token Attribution:** Look at these weight badges below: unauthorized (+0.24), never made (+0.21), unauthorized charge (+0.21).*

***Why is this critical for regulatory compliance?** Banking compliance officers strictly reject black-box deep learning models where you cannot justify why an automated classification occurred. Our model computes exact linear feature attribution: **z_c = b_c + SUM(w_{c,i} * x_i)**. Every word is scored individually based on its trained linear hyperplane coefficient w_{c,i} and TF-IDF feature weight x_i, providing auditable mathematical proof for regulatory compliance."*

Step 4: Complaint Explorer & Slide-Out Audit Drawer [5:00 – 6:30]

[Action]: Click 'Complaint Explorer' in top navbar. Click preset pill '■ UPI & Transaction Outage'. In search box, type 'escrow'. Click any row to open the slide-out drawer.
*"Next is the **Complaint Explorer**, designed for compliance officers handling dispute audits.*
*• **One-Click Analytical Presets:** With a single tap, an auditor can isolate 'UPI & Transaction Outage' (failed transfers), 'ATM Skimming & Fraud' (compromised cards), or 'Overdraft / Double-Debit' (billing calculation errors).*
*• **Sub-10ms Debounced Search:** Notice that as I type 'escrow', our sub-10 millisecond debounced full-text search filters across all 6,924 complaint records in real time with zero UI lag.*

• ***Slide-Out Inspection Drawer:*** *Clicking any complaint record opens our Slide-Out Inspection Drawer. Without losing search context, an auditor can examine the customer's verbatim narrative, product vertical (Home Mortgage, Credit Card, Savings Account), submission channel (Mobile App, Net Banking, Branch, Ombudsman), and model-assigned dispute tags."*

Step 5: NLP Intelligence — Model Benchmarks, Calibration & ECE [6:30 – 8:00]

[Action]: Click 'NLP Intelligence' in top navbar. Point to the Model Comparison Table, highlighting Linear SVM vs Logistic Regression vs Baseline. Scroll to Confusion Matrix and Per-Class metrics.

*"Under the **NLP Intelligence** tab, we present our supervised model evaluation on held-out test splits:*

• ***Benchmark Comparison:*** *We benchmarked multiple classifiers on sublinear TF-IDF features: Majority Baseline scored ~18.5%, Logistic Regression reached ~92.4%, and our **Calibrated Linear Support Vector Machine achieved 98.7% Accuracy and 0.985 Macro F1 score**.*

• ***Why Linear SVM over Deep Neural Networks?*** *In text classification with a 10,000-feature unigram/bigram vocabulary, the data space is high-dimensional and sparse. SVM's convex hyperplane margin maximization provides superior separation compared to tree models, while executing inference in under 2 milliseconds on standard CPU hardware with zero GPU dependency.*

• ***Platt Sigmoid Calibration & ECE:*** *Notice the **ECE (Expected Calibration Error)** column. A standard SVM outputs uncalibrated geometric distances from the hyperplane. We applied Platt Scaling—fitting a cross-validated logistic sigmoid—guaranteeing that an output confidence of 95% represents a true 95% empirical probability.*

• ***Confusion Matrix & Influential Terms:*** *Below, the confusion matrix heatmap confirms near-perfect diagonal concentration across all 8 classes, while the Influential Terms section displays the top learned positive coefficients for each banking category."*

Step 6: ML Intelligence — Topics, Rolling MAD & Holt-Winters [8:00 – 9:45]

[Action]: Click 'ML Intelligence' in top navbar. In 'Clustering' tab, show 2D PCA cluster map and silhouette table. Switch to 'Anomalies' tab. Switch to 'Forecasting' tab and toggle 7/14/30 day horizon.

*"Now we move to **ML Intelligence**, which houses our unsupervised discovery and time-series algorithms:*

*1. **Unsupervised Latent Topics (Tab 1):** Supervised learning only categorizes known classes. Using **Non-Negative Matrix Factorization (NMF, k=8)** and **K-Means Clustering (k=5)**, the system automatically extracts emerging complaint themes—like ATM skimming surges or escrow tax delays—without needing human labels. We scanned k from 3 to 8, choosing k=5 by maximizing the silhouette coefficient, and mapped them in 2D using PCA.*

*2. **Seasonality-Adjusted Anomaly Detection (Tab 2):** In banking, complaint volume drops by 70% every Saturday and Sunday. A standard 3-sigma Gaussian test triggers false alarms every Monday and misses real weekend spikes. We formulated a **Day-of-Week Normalized Rolling MAD Algorithm**: we compute the median and Median Absolute Deviation over a 28-day window partitioned by day-of-week. Because median and MAD have a 50% breakdown point, historical surges do not corrupt the baseline, eliminating 100% of weekend false alarms.*

*3. **Capacity Planning via Damped Holt-Winters (Tab 3):** To prevent regulatory SLA breaches, we forecast incoming complaint volume 14 to 30 days in advance using Additive Damped Holt-Winters Exponential Smoothing. It tracks level (l_t), trend (b_t), and 7-day cyclical weekly seasonality (s_t), with a damping parameter phi = 0.90 to prevent linear runaway projections. Evaluated using expanding-window chronological backtesting, it reduced forecasting error by over 32% compared to standard baselines."*

Step 7: Alert Center & Incident Deduplication [9:45 – 10:45]

[Action]: Click 'Alert Center' in top navbar. Point to active alerts and priority badges. Toggle status from 'TRIGGERED' -> 'INVESTIGATING' -> 'RESOLVED'. Point to root-cause investigation notes.

*"Our final operational module is the **Alert Center**. In real production monitoring, anomaly detectors suffer from **Alert Fatigue**: a multi-day system outage floods teams with hundreds of duplicate tickets.*

*To prevent this, we engineered **Deterministic Deduplication Keying**: hash(product, anomaly_type, date_bucket). When a spike persists, it updates the existing incident timeline instead of spamming analysts with new alerts.*

*Analysts follow an auditable state machine: **TRIGGERED -> INVESTIGATING -> RESOLVED**. Analysts can record investigation notes and tag verified root causes (such as 'Core Banking Firmware Upgrade' or 'UPI Switch Timeout'), creating an immutable audit trail stored in SQLite for regulatory review."*

Step 8: Conclusion & Individual Team Contributions [10:45 – 11:30]

[Action]: Navigate back to Home Command Center. Summarize outcomes cleanly and open for questions.

"To conclude our presentation:

• ***98.7% classification accuracy** across 8 banking verticals with calibrated probabilities and token attribution.*

*• **Sub-20ms API latency** using an asynchronous FastAPI architecture.*
*• **Zero false positive weekend alerts** using Day-of-Week normalized Rolling MAD.*
*• **32% forecast error reduction** via Damped Holt-Winters exponential smoothing.*

Individual Team Contributions:
*• **Aryan Turalkar (A152):** Designed NLP preprocessing, sublinear TF-IDF vectorization, LinearSVM training, Platt scaling calibration, and feature attribution.*
*• **Suryansh Saraf (A-125):** Architected the full-stack system, designed the React 18 / Tailwind Command Center, built the FastAPI REST services, and set up deployment.*
*• **Soham Hande (A-129):** Formulated the Day-of-Week Rolling MAD algorithm, integrated the multivariate Isolation Forest ensemble, and conducted NMF latent topic modeling.*
*• **Pratham Deepak (A-114):** Implemented the Damped Holt-Winters time-series model, expanding-window walk-forward backtesting, and built the Alert Center deduplication lifecycle.*

Thank you, professors. We are now open for your questions and discussion."

3. Course Policy (702AI0C009) Curriculum Mapping & Theory Deep-Dive

The Natural Language Processing course policy (702AI0C009) at MPSTME specifies 4 Course Outcomes (CO1–CO4), 7 Theory Units, and 10 Lab Experiments. ComplaintIQ addresses all course outcomes with rigorous mathematical implementations:

CO1 / Unit 1 & 2: Text Preprocessing & Negation Preservation (Exp 1, 2)
• **Contraction Expansion & Cleaning:** Regular expressions expand contractions (won't $\to$ will not, cannot $\to$ can not) and strip non-alphabetic noise while preserving word boundaries.
• **The Negation Preservation Trick:** Standard stopwords (NLTK) blindly discard words like not, no, never, without. In banking complaints, discarding negations inverts sentence polarity (e.g., 'cash **not** dispensed' becomes 'cash dispensed'). ComplaintIQ explicitly isolates negations in a protected set, allowing them to bind to adjacent verbs in bigrams.

CO2 / Unit 2 & 3: Sublinear TF-IDF & (1,2)-Gram Language Modeling (Exp 3, 5)
• **Mathematical Formulation:** Feature representation utilizes sublinear term frequency: **tf_{sublinear} = 1 + log(tf)** for tf > 0, multiplied by Inverse Document Frequency **idf(t) = log((1 + n) / (1 + df(t))) + 1**.
• **N-Gram Vocabulary:** Extracts unigrams and bigrams (e.g., unauthorized charge, late fee, minimum balance), capturing critical compound expressions up to 10,000 dimensions with sublinear scaling to prevent repetitive legal boilerplate from dominating vectors.

CO2 / Unit 4: Syntax Level Analysis & Linear Feature Attribution (Exp 6, 7)
• **Decision Logit Equation:** Local feature attribution for class c is computed as: **z_c = b_c + SUM(w_{c,i} * x_i)**, where w_{c,i} is the learned hyperplane weight and x_i is the TF-IDF feature value.
• **White-Box Auditability:** Every token's contribution is isolated: positive weights provide affirmative evidence, while negative weights penalize alternative categories. Pre-computed weights allow local explanations in under 2ms without slow surrogate approximations like LIME or SHAP.

CO3 / Unit 5: Semantic Analysis, Latent Topics & Clustering (Exp 4, 8, 9)
• **Non-Negative Matrix Factorization (NMF, k=8):** Decomposes term-document matrix **V_{N x M} approx W_{N x K} * H_{K x M}** into 8 latent topics. The non-negativity constraint guarantees purely additive, parts-based semantic representations without negative cancellations.
• **K-Means Clustering with Silhouette Optimization:** Scans cluster counts k from 3 to 8, evaluating the Silhouette Coefficient: **s(i) = (b(i) - a(i)) / max(a(i), b(i))** to select the optimal natural cohort grouping (k=5), visualized via 2D PCA projection.

CO4 / Unit 7: Multi-Class Text Classification, Sentiment & Platt Scaling (Exp 10, 11)
• **Calibrated Linear SVM:** Solves convex margin optimization: min 0.5 * ||w||^2 + C * SUM(xi_i).
• **Platt Sigmoid Calibration:** Fits a logistic sigmoid **P(y=1 | f) = 1 / (1 + exp(A*f + B))** via maximum likelihood on cross-validation folds, transforming uncalibrated margin distances into compliant posterior probabilities and evaluated using Expected Calibration Error (ECE).

4. Frontend UI Visual Map (Where Every Concept Lives on Screen)

Frontend Page & Component	NLP / ML Concept Implemented	Visual Indicators on Screen
Home: Live AI Classifier (LivePredictBox.tsx)	• Sublinear TF-IDF Vectorization • Calibrated Linear SVM • Platt Scaling ($P \in [0, 1]$) • Token Attribution ($w_i \cdot x_i$) • Sentiment Polarity	• Result rendered in < 20ms • **99.9% Confidence** badge • Alternative category probability bars • **Influencing term badges** with +weights • Sentiment score and colored balance meter
Home: Command Center (Overview.tsx)	• Day-of-Week Rolling MAD Anomaly Surveillance • Category Distribution • Channel Ingestion Mix	• 4 Top KPI cards (Volume, Negative %, Sentiment, Warnings) • Recharts 30-day dual line trajectory • **Red circular surge anomaly markers** • Categories bar chart & Sentiment donut chart
Complaint Explorer (Explorer.tsx)	• Sub-10ms Debounced Full-Text Search • Curated Operational Presets • Contextual Narrative Inspection	• Search bar with real-time text debouncing • 4 Preset pills (Overdraft, App Login, UPI, Skimming) • Multi-column filter dropdowns • **Slide-out inspection drawer** showing full complaint
NLP Intelligence (NLPIntelligence.tsx)	• Supervised Model Benchmarking • Expected Calibration Error (ECE) • 5-Fold Cross Validation • Confusion Matrix & Macro F1	• Model comparison table (SVM deployed vs LR vs Baseline) • **Confusion matrix heatmap** (diagonal concentration) • Per-class Precision, Recall, F1 table • **Influential terms per class** keyword cards
ML Intelligence: Clusters (MLIntelligence.tsx)	• Latent Topic Modeling (NMF, k=8) • K-Means Document Clustering • Silhouette Score Optimization • 2D PCA Dimensionality Reduction	• **2D PCA Cluster Scatter Plot** • Silhouette score evaluation scan table (k=3 to 8) • Cluster cards with extracted semantic topics, percentage share, and representative text snippets
ML Intelligence: Forecast (MLIntelligence.tsx)	• Damped Additive Holt-Winters Exponential Smoothing (m=7, phi=0.90) • Expanding-Window Walk-Forward Validation	• 7/14/30-Day forecast trajectory chart with confidence bands • Holdout evaluation table comparing Damped Holt-Winters vs Seasonal Naive vs Rolling Mean (MAE, RMSE, Bias)
Alert Center (AlertCenter.tsx)	• Deterministic Deduplication Hashing • Incident State Lifecycle Machine • Regulatory Root-Cause Ledger	• Active incident ticket cards with CRITICAL/WARNING badges • **hash(product, anomaly, date)** deduplication key • **Status toggle: TRIGGERED -> INVESTIGATING -> RESOLVED** • Root-cause notes and timestamped audit log

5. Faculty Viva & Cross-Examination Defense (Top 10 Lethal Questions)

Be fully prepared for examiners challenging your algorithmic and architecture decisions. Here are the exact answers to the 10 most common and difficult questions faculty ask:

Q1: Why didn't you use BERT, RoBERTa, or a Generative LLM for text classification?

Defense: In production banking compliance, sub-20ms latency and exact token attribution are legally mandated. Linear SVM runs in under 2ms with zero GPU dependency, achieves 98.7% accuracy, and provides exact mathematical weights (w_i * x_i) that compliance auditors can verify, unlike black-box deep networks requiring heuristic approximations like LIME.

Q2: Why use modified Z-score with MAD instead of standard 3-sigma standard deviation?

Defense: Standard deviation uses the mean and variance, which are distorted by extreme spikes (outlier masking). The median and MAD have a 50% breakdown point, resisting outlier contamination. Combining MAD with 28-day Day-of-Week grouping accounts for the 70% weekend volume drop and eliminates false Monday alarms.

Q3: What is Platt scaling and why couldn't you just use SVM's decision function?

Defense: A Support Vector Machine outputs an uncalibrated signed hyperplane distance f(x) in (-inf, +inf). While the sign gives the class, the magnitude is not a probability. Platt scaling fits a logistic sigmoid P(y=1 | f) = 1 / (1 + exp(A*f + B)) via maximum likelihood over cross-validation folds, guaranteeing that a 95% confidence score represents a true 95% empirical likelihood.

Q4: How did you prevent data leakage during time-series forecasting validation?

Defense: Standard k-fold cross-validation or random train-test splitting introduces lookahead bias because future points bleed into past training splits. We implemented Expanding-Window Chronological Backtesting: the model was trained strictly on historical data up to day T, and evaluated on unseen future days T+1 to T+14.

Q5: Why did you use NMF instead of LDA (Latent Dirichlet Allocation) for topic modeling?

Defense: LDA is sensitive to Dirichlet prior hyperparameters (alpha, beta) and frequently generates overlapping, noisy topics on short customer texts. NMF uses non-negative matrix decomposition (V approx W * H). Because negative values are forbidden, features can only be added together, creating clean, interpretable, parts-based topics for business teams.

Q6: What is the purpose of the damping parameter (phi = 0.90) in Holt-Winters?

Defense: Standard Holt-Winters assumes trend continues linearly to infinity, leading to unrealistically steep forecasts over 14–30 days. The damping factor (phi = 0.90) damps trend accumulation: SUM(phi^i * b_t), leveling off projections to reflect realistic banking operational capacity.

Q7: How does the system handle alert fatigue in the Alert Center?

Defense: If an infrastructure outage lasts 4 days, naive monitors fire hundreds of duplicate emails. We implemented deterministic hashing: dedup_key = hash(product, anomaly_type, date_bucket). When a spike persists, the system increments the incident count and updates the existing ticket rather than creating redundant alerts.

Q8: How does your system perform local feature attribution?

Defense: Because our features are normalized TF-IDF weights x_i and our classifier is linear with learned weights w_{c,i}, the classification logit is z_c = b_c + SUM(w_{c,i} * x_i). Every token's contribution is isolated and computed in under 2 milliseconds without needing perturbation-based surrogate models.

Q9: What happens if a complaint talks about multiple issues simultaneously?

Defense: Our probability calibration handles this cleanly. Instead of just returning a single hard label, the API returns the full posterior probability distribution across all 8 classes. If a customer mentions an ATM cash failure and an overdraft penalty, the model outputs, for example, 55% Transaction Failure and 38% Billing & Fees, highlighting tokens from both.

Q10: How would this system scale if complaint volume increased from 7,000 to 1,000,000?

Defense: The FastAPI backend is stateless and horizontally scalable behind NGINX or Kubernetes. TF-IDF vectorizers and SVM models are serialized pickle artifacts loaded into RAM once at startup, serving predictions in under 2ms. The SQLite store can be transitioned seamlessly to PostgreSQL or Amazon RDS with zero application code changes.

6. Individual Contribution Defenses (What to Say When Asked: 'What Did YOU Do?')

Aryan Turalkar (Roll No: A152) — NLP & Machine Learning Lead

*"I designed and implemented the Natural Language Processing and supervised classification core. Specifically, I engineered the custom regex preprocessing pipeline, contraction expansion, and negation-preserving tokenizer. I tuned the sublinear TF-IDF vectorizer over 10,000 unigrams and bigrams, trained the LinearSVC model achieving 98.7% accuracy, implemented Platt sigmoid probability calibration with ECE evaluation, and formulated the linear feature attribution algorithm (w_i * x_i) for white-box interpretability."*

Suryansh Saraf (Roll No: A-125) — Full-Stack Architect & UI/UX Lead

"I architected the entire full-stack system end-to-end. I built the modern React 18, TypeScript, and TailwindCSS Command Center, Complaint Explorer with sub-10ms debounced search, and the interactive Live Predictor widget. On the backend, I engineered the asynchronous FastAPI REST microservices, database schemas in SQLAlchemy/SQLite, and edge deployment configurations, ensuring sub-20ms round-trip latency."

Soham Hande (Roll No: A-129) — Statistical Anomaly Detection & Clustering Lead

"I developed the statistical surveillance and unsupervised discovery engines. I formulated the Day-of-Week normalized Rolling MAD algorithm that groups data by weekday to eliminate weekend volume bias, integrated the multivariate Isolation Forest ensemble, and tuned the multi-tier severity thresholds (Info, Warning, Critical). Furthermore, I implemented Non-Negative Matrix Factorization (NMF, k=8) and K-Means clustering with silhouette score optimization."

Pratham Deepak (Roll No: A-114) — Time-Series Forecasting & Operations Lead

"I led the predictive time-series modeling and incident operational workflow. I formulated the Additive Damped Holt-Winters exponential smoothing model with 7-day seasonality, performed expanding-window chronological backtesting demonstrating a 32% error reduction over baseline models, and engineered the Alert Center deterministic deduplication keying and incident lifecycle state machine."

EXAMINATION CONFIDENCE TIP

Final Presentation Advice: Stay calm, speak clearly, and let the live working frontend do the talking. Whenever a faculty member asks a conceptual question, use the navbar to switch to the relevant tab (NLP Intelligence, ML Intelligence, Explorer) and point directly to the numbers and graphs. Good luck!